

WAKE FIELDS AND IMPEDANCES

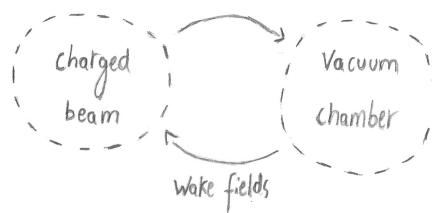
Notes from:

Henrique W. Silvestre

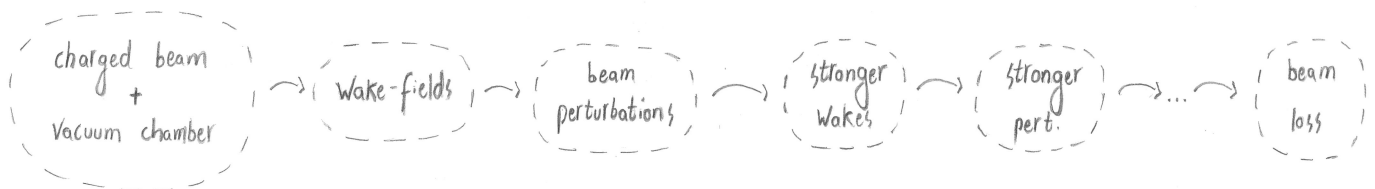
1 - Wake Fields and Impedance

1.1 - Introduction

When a beam of charged particles travels inside the metallic vacuum chamber of a particle accelerator, its moving electromagnetic field induces charges and currents in the surface of the chamber's walls - which "seek" to cancel the fields inside the walls. These induced moving charges, in turn, generate their own EM fields, the so-called wake fields, which will then interact with the charged beam itself. This interaction can lead to disturbances in the beam, which will translate into changes in the wake fields, and the cycle repeats.



The main problem arises if the conditions are such that the effect of the wake fields over the beam is to create perturbations which intensify the wake fields themselves, then intensifying the perturbations, and so on. This feedback mechanism can ultimately lead to beam instabilities - in this context, called collective instabilities - and thus to a subsequent beam loss.



collective instability mechanism

With this in mind, the critical and ever-growing importance of understanding and curing these mechanisms in particle accelerators becomes evident → with the strive of achieving higher and higher beam intensities and brightness, which imply greater beam currents, the use of ID's, and narrower vacuum chambers (to enable imposing stronger mag. fields on the beam), the strength and thus impact of wake fields also gets more and more substantial.

Having established the basics of what are wake fields and why they are important, let's now get a better physical, intuitive understanding of why - and in which conditions - these fields appear.

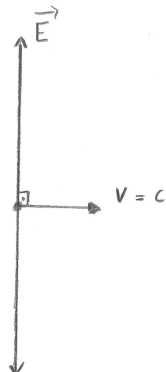
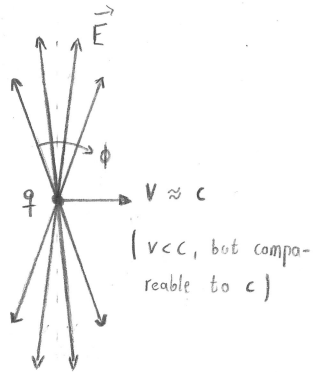
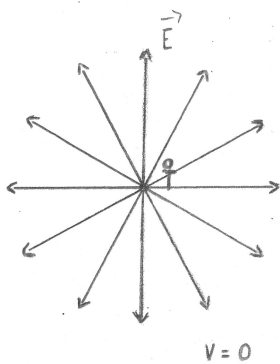
* The following subsections were based mainly on [Chao, OCPA 2010], [Stupakov, 2000] and [CERN-Accel-School, 2017], and will adopt SI units!

1.1.1 - Moving Charges in Free Space

First of all, let's consider the electric field lines of an isolated electron.

→ Gauss Law, $\vec{\nabla} \cdot \vec{E} = \rho/\epsilon_0$, shows that a charged particle is always attached to electric field lines, regardless of the circumstances (one cannot "shake off" these lines).

→ For a stationary charge in free space, these lines radiate radially (outwards or inwards depending on the sign of the charge) isotropically.



→ If the charge has a velocity v near the speed of light, the field lines get Lorentz-contracted into a thin "disk"/"pancake" perpendicular to the direction of motion, with an opening angle in the order of $1/\gamma$ ($\gamma = (1 - v^2/c^2)^{-1/2}$).

if Sirius: $\gamma \approx 5870.85$
 $\Rightarrow \phi \approx 1/\gamma \approx 170 \mu\text{rad}$

→ For an ultrarelativistic charge, $v = c$, and the pancake reduces to an infinitely thin sheet.

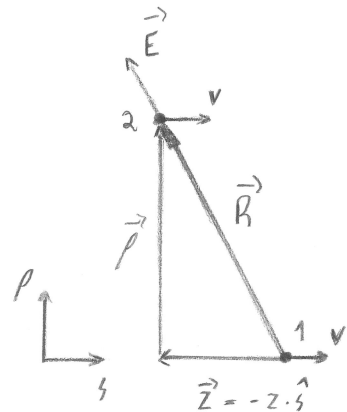
↳ Since a moving charge is an electric current, it will generate a magnetic field $\vec{B} = \frac{1}{c^2} \vec{v} \times \vec{E}$

↳ Consequently, all electromag. field lines are contained in a plane transverse to $\vec{v}$ ($\vec{E} = E \hat{\rho}$, $\vec{B} = B \hat{\phi}$) in the ultrarel. limit

In a bit more detail, consider a source particle of charge q_1 moving with constant velocity $\vec{v} = v \cdot \hat{z}$, and a trailing particle of charge q_2 and the same velocity, at a position $\vec{R} = \vec{\rho} - z \hat{z}$ w.r.t. particle 1 - long. dist. z behind and w/ a transv. disp. $\vec{\rho}$.

Then, defining $R_*^2 = z^2 + \rho^2/\gamma^2$, the EM fields from q_1 experienced by q_2 will be [Stupakov, 2000]:

$$\vec{E} = \frac{q_1}{4\pi\epsilon_0} \frac{\vec{R}}{\gamma^2 R_*^3} ; \quad \vec{B} = \frac{1}{c^2} \vec{v} \times \vec{E}$$



Furthermore, the resulting Lorentz force experienced by q_2 , $\vec{F} = q_2 (\vec{E} + \vec{v} \times \vec{B})$ will have the following longitudinal and transverse components:

$$F_{||} = -\frac{q_1 q_2}{4\pi\epsilon_0} \frac{z}{\gamma^2 (z^2 + \rho^2/\gamma^2)^{3/2}} ; \quad \vec{F}_{\perp} = \frac{q_1 q_2}{4\pi\epsilon_0} \frac{\vec{\rho}}{\gamma^4 (z^2 + \rho^2/\gamma^2)^{3/2}}$$

From this, one can see that, for $z \neq 0$, $F_{||} \propto \gamma^{-2}$ and $\vec{F}_{\perp} \propto \gamma^{-4}$, meaning that $\vec{F}$ - the so-called space-charge force - tends to zero as $\gamma \rightarrow \infty$ ($v \rightarrow c$).

Moreover, in the ultrarelativistic limit the fields become:

$$\vec{E} = \frac{q_1}{2\pi\epsilon_0 \rho} \delta(z-ct) \hat{\rho} ; \quad \vec{B} = \frac{q_1}{2\pi\epsilon_0 c \rho} \delta(z-ct) \hat{\phi}$$

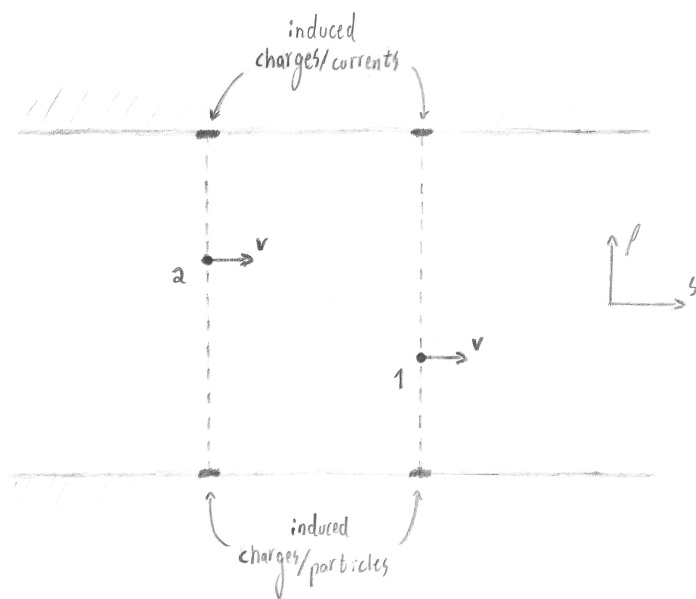
And thus $\vec{F} = \vec{F}_{\perp}$, which as we've seen decreases with γ as γ^{-4} for $z \neq 0$ and as γ^{-1} for $z = 0$, meaning that $\vec{F}$ completely vanishes when $v \rightarrow c$.

In other words, the electric and magnetic forces perfectly compensate each other for moving charges in free space in the ultrarel. limit, such that the electromagnetic interaction disappears.

1.1.2 - Moving Charges in a Perfectly Conducting Pipe

Now, consider the same charges from before, but moving inside a perfectly conducting cylindrical pipe, parallel to its axis. The moving charges will induce charges and currents on the surface of the wall, which shields the metal from the EM field of the particles, perfectly cancelling out the fields inside the walls.

In this precise context, the corresp. image charges travel with the same velocity v of the particles, and on parallel paths. Consequently, in the limit $v \rightarrow c$, according to the previous results, none of the charges (image or not) interact with each other - no matter how close to the wall the particles are.



Notice that this last remark remains valid even if the pipe cross-section is not circular. In fact, the only needed condition (geometrically) is that the cross-section shape does not vary with s .

Thus, for ultrarelativistic charged particles moving inside a perfectly conducting pipe of cte. cross-section, there is no net electromagnetic interaction between the particles, and in particular, no EM fields are left behind them - i.e., there are no wake fields!

1.1.3 - Origins of the wake fields

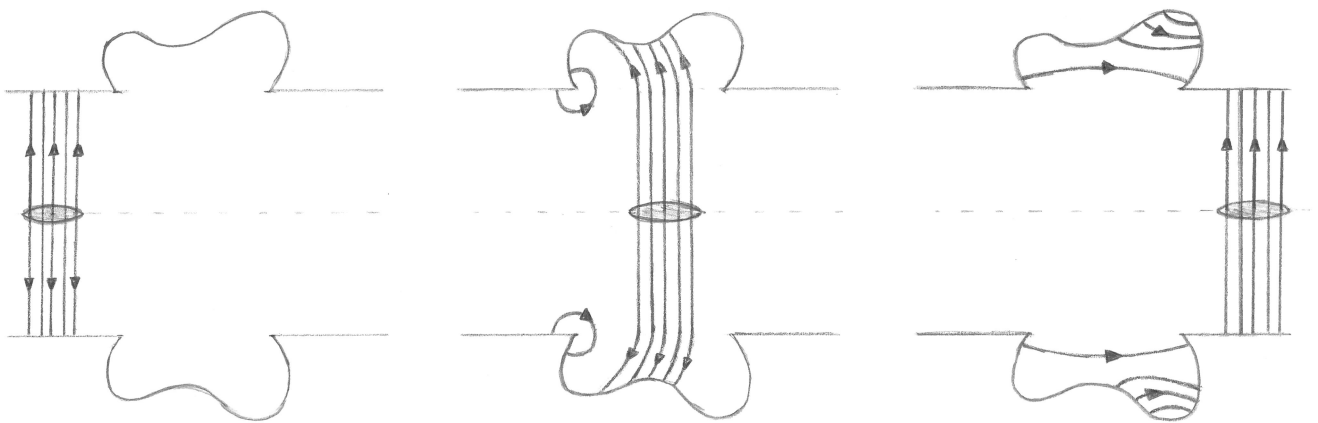
Up until now, we've seen only the ideal scenarios where no net EM interaction between the particles happen, and no wake fields are generated. However, one can probably already intuit which conditions, when broken, will lead to the formation of wake fields.

Considering the ultrarelativistic limit - meaning that space-charge forces can be neglected - the two main origins of wake fields are: (i) the chamber's cross-section changes with s ; (ii) the chamber's material is not a perfect conductor.

Let's get a bit more intuition on how the wake fields are formed in each of these cases.

(i) Geometric Discontinuities

When a charged particle beam traverses a discontinuity in the vacuum chamber - meaning a variation in the chamber's cross-section - the image charges have their trajectories bent, experiencing an acceleration and thus radiating. In this sense^(and context), the wake fields can be understood heuristically as the radiation fields left behind by the image charges when their trajectories are bent! [Chao, OCPA 2010]

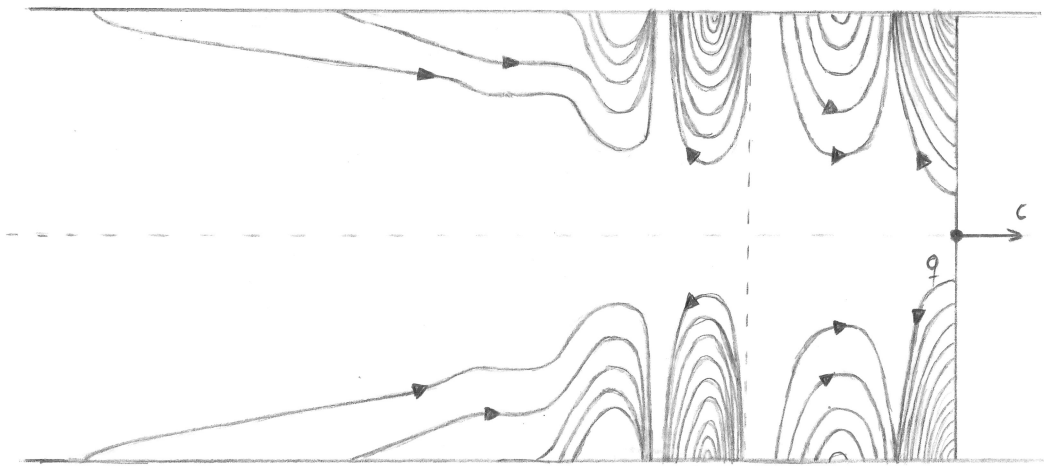


Electric field lines and wake fields from a charged bunch passing through a perfectly conducting chamber with a cavity [Chao, OCPA 2010]

(ii) Resistive Wall

When the chamber walls are not perfectly conducting, but rather have a finite conductivity (σ_{el}) - i.e. a non-zero resistivity (ρ_{el}) - the ^{induced} image charges and currents cannot follow the beam perfectly anymore. Instead, their response is delayed and part of the beam energy is dissipated as heat in the walls. As a result, EM fields remain behind the source, giving rise to the so-called resistive-wall wake fields.

An example of RW wake fields left behind by a charge q is illustrated below:



[Chao, OCPA 2010]